CampSpark – Smart College Management System

Complete System Design & Module Details (Final Integrated Version)

🔐 CORE DESIGN PRINCIPLES

1. Role-Based Access Control (RBAC)
2. Department-Based Data Isolation
3. Purpose-Based Fee & Data Visibility
4. AI as Assistant, NOT Decision Maker
5. Multi-Level Verification for Sensitive Actions
6. End-to-End Privacy & Audit Logging

📦 MODULES OVERVIEW

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1. Authentication & Role Management Module

Roles Supported:

- Student
- Faculty
- Admin
- NSS Incharge
- Club Incharge
- Event Incharge
- Complaint Officer (Male / Female)
- Verifier Admin (Principal / Higher Authority)

Features:

- Secure login
- Role assignment
- Department mapping
- Permission enforcement

Rule:

Every user sees **only what their role allows**.

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2. Student Module (Strict Privacy)

Features:

- View personal profile
- View attendance
- View marks & results
- View own fee status only
- Submit complaints
- Receive private notifications
- View today dashboard & reminders

Restrictions:

- No AI access
- No other student data
- No analytics

Privacy Rule:

Each student can access **only their own data**.

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3. Faculty Module (Department-Scoped)

Features:

- View & manage students of own department only
- Attendance management
- Marks entry & review
- Academic performance tracking
- Department-level reports
- Timetable & exam handling
- Daily task dashboard

AI-Assisted Capabilities (Limited):

- Weak student identification (own dept only)
- Attendance trend analysis
- Performance summaries

Restrictions:

- No other department data
- No fee access
- No complaint decisions

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4. Course & Academic Management Module

- Course creation
- Subject management
- Semester & credit structure
- Academic calendar
- Syllabus upload

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5. Attendance Management Module

- Daily attendance
- Period-wise & subject-wise attendance
- Attendance percentage calculation
- Low attendance detection
- Attendance reports

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6. Exam & Evaluation Management Module

- Exam creation & configuration
- Internal, model, lab, semester exams
- Faculty-controlled exam add / modify / remove
- Exam scheduling & rescheduling

- Marks entry & validation
- Subject-wise marks listing
- Grade auto calculation
- GPA & CGPA calculation
- Rank list generation
- Student-profile integrated marks view
- AI-based weak student detection (assistive only)

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7. Notification & Communication Module

- Text, voice & video announcements
- Circulars
    - College-wise
    - Department-wise
    - Class-wise
    - Year-wise
    - Student-wise
- Event communication
    - Upcoming events (what, when, where)
    - Event history (photos, programs, winners, prizes)
- Email alerts
- SMS alerts
- In-app notifications
- Read tracking & notification history

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8. Fee Management & Financial Support Module (Scoped Access)

Fee Types:

- Academic Fee
- NSS Fee

- Club Fee
- Event Fee

Access Rules:

- NSS Incharge → NSS Fee only
- Club Incharge → Club Fee only
- Event Incharge → Event Fee only
- Admin → All fees
- Faculty → No fee access

Features:

- Fee structure management
- Result-linked fee logic
- Scholarship management
    - Tamil Pudhalvan
    - Pudhumai Pen
    - BC, MBC, SC, ST, Minority
    - Special quota
    - Central Government
    - Merit, Sports, Cultural, EWS, First Graduate
- Scholarship eligibility rule engine
- Scholarship approval workflow
- Sponsor management
- Special quota fee concessions
- EMI & installment settlement
- Fee notifications
- Student fee dashboard

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9. Timetable Management Module

- Class, faculty, subject & lab timetables

- Faculty availability & preference
- Auto timetable generation
- Clash detection
- Extra & remedial classes
- Exam & event sync
- Substitute faculty management
- Student & faculty views
- Timetable update notifications
- Timetable history & version control
- Attendance synchronization
- Holiday calendar integration
- Faculty workload analytics

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10. Document Management Module

- Student documents
- Faculty & staff documents
- Academic documents
- Administrative documents
- Event photos & reports
- Scholarship & fee documents
- Upload & version control
- Verification & approval workflow
- Role-based access
- Search & download
- Backup & security

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11. Report Generation Module

- Student academic reports

- Attendance reports
- Fee & scholarship reports
- Exam & result analytics
- Faculty performance reports
- Department & college summary reports
- Event & activity reports
- Custom report builder
- Visual charts
- PDF & Excel export
- Scheduled auto reports

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12. Complaint & Grievance Module (High Security)

Complaint Types:

- Academic
- Harassment
- Infrastructure
- Fee-related
- Administrative

Complaint Committee:

- Admin
- Male Complaint Officer
- Female Complaint Officer

AI Role:

- Categorization
- Summarization
- Pattern detection
- Priority suggestion

Complaint Status Flow:

Submitted → Under Review → Under Investigation → Resolved / Rejected

Each change:

- Requires reason
- Logged permanently
- Student notified privately

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13. Special Roles Module

Common Rules:

- Access only assigned activity data
- No academic data
- No global student data

NSS Module

- NSS students
- Activity records
- NSS fund / fee
- Participation reports
- AI summaries (non-personal)

Club Module

- Club members
- Club events
- Club fees
- Participation records
- AI club reports

Event Module

- Event registration
- Event fees
- Participation
- Expense tracking

- AI event summaries

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14. Daily Dashboard & Smart Reminder Module

- Role-based today dashboard
- Today timetable
- Today exams & tests
- Today tasks & approvals
- Today events
- One-time per day reminder logic
- Home screen mini widget
- Reminder history

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15. Admin Dashboard & System Control Module

- College-wide analytics
- Department analytics
- User & role management
- Fee oversight
- Complaint oversight
- AI insights (aggregate only)

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16. Verification & Second Opinion Module

Sensitive Actions:

- Complaint resolution / rejection
- Fee refunds
- Disciplinary actions
- Critical data modifications

Workflow:

Admin Action → Pending Verification → Verifier Admin Approval → Final Execution

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17. Audit & Logging Module

Logged Actions:

- Complaint access & updates
- Fee updates
- Admin decisions
- Verification approvals

Log Rules:

- Read-only
- Non-deletable
- Tamper-proof

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18. AI Integration Module (LIMITED & SAFE)

AI Access:

- Faculty → own department only
- Admin → aggregate data only
- Special roles → summary only

AI Restrictions:

- No student identity exposure
- No decision automation
- No cross-role data access

🧠 **SYSTEM ARCHITECTURE FLOW**

User Login
→ Role & Department Check
→ Module-Level Access Control
→ AI Helper Layer (Restricted)
→ Filtered Database
→ Audit Log + Private Notification

✅ **FINAL SYSTEM GUARANTEES**

- Strong privacy protection
- Government-level security design
- AI used only as assistant
- Zero unauthorized data exposure
- Fully viva-ready architecture